

Proof. Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be real numbers.

1. A non-negative quadratic. For a real parameter t define

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2.$$

Each summand is a square, hence $f(t) \geq 0$ for all $t \in \mathbb{R}$.

2. Expansion of $f(t)$. Using $(a_i - t b_i)^2 = a_i^2 - 2t a_i b_i + t^2 b_i^2$, we obtain

$$\begin{aligned} f(t) &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= C - 2Dt + Bt^2, \end{aligned}$$

where

$$C = \sum_{i=1}^n a_i^2, \quad D = \sum_{i=1}^n a_i b_i, \quad B = \sum_{i=1}^n b_i^2 \geq 0.$$

3. Discriminant argument. If $B > 0$, the quadratic $Bt^2 - 2Dt + C$ is non-negative for every real t . For a quadratic $at^2 + bt + c$ with $a > 0$ this is equivalent to its discriminant being non-positive:

$$\Delta = (-2D)^2 - 4BC = 4(D^2 - BC) \leq 0 \implies D^2 \leq BC.$$

Substituting the definitions of B, C, D yields the Cauchy–Schwarz inequality

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

If $B = 0$ then all $b_i = 0$ and the inequality is trivial.

4. Equality case. Equality $D^2 = BC$ occurs exactly when $\Delta = 0$, i.e. when the quadratic has a real root. Since $f(t) \geq 0$ and $f(t) = 0$ at that root, we must have

$$f(t_0) = 0 \quad \text{for} \quad t_0 = \frac{D}{B} \quad (B \neq 0).$$

Thus

$$0 = f(t_0) = \sum_{i=1}^n (a_i - t_0 b_i)^2,$$

which forces each summand to vanish: $a_i = t_0 b_i$ for all i . Hence equality holds iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (or trivially when one of the vectors is the zero vector).

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